

4.13.13

EE25BTECH11008 - Anirudh M Abhilash

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Question

If one of the lines of

$$my^2 + (1 - m^2)xy - mx^2 = 0$$

is a bisector of the angle between the lines

$$xy = 0,$$

then find the value of m .

Solution

The given homogeneous equation can be expressed as

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0. \quad (1)$$

Let

$$A = \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix}.$$

Each line represented by this homogeneous equation passes through the origin, and its direction vector $\mathbf{v}$ satisfies

$$\mathbf{v}^T A \mathbf{v} = 0. \quad (2)$$

For the bisectors of the coordinate axes ($xy = 0$), the directions are along $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
Substituting $\mathbf{v}_1$,

$$\mathbf{v}_1^T A \mathbf{v}_1 = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (3)$$

$$= -m + (1 - m^2) + m = 1 - m^2. \quad (4)$$

Similarly, substituting $\mathbf{v}_2$,

$$\mathbf{v}_2^T A \mathbf{v}_2 = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (5)$$

$$= -m - (1 - m^2) + m = m^2 - 1. \quad (6)$$

For either $\mathbf{v}_1$ or $\mathbf{v}_2$ to represent a line of the pair,

$$1 - m^2 = 0. \quad (7)$$

Hence,

$$m = \pm 1. \quad (8)$$

$$\boxed{m = \pm 1}$$

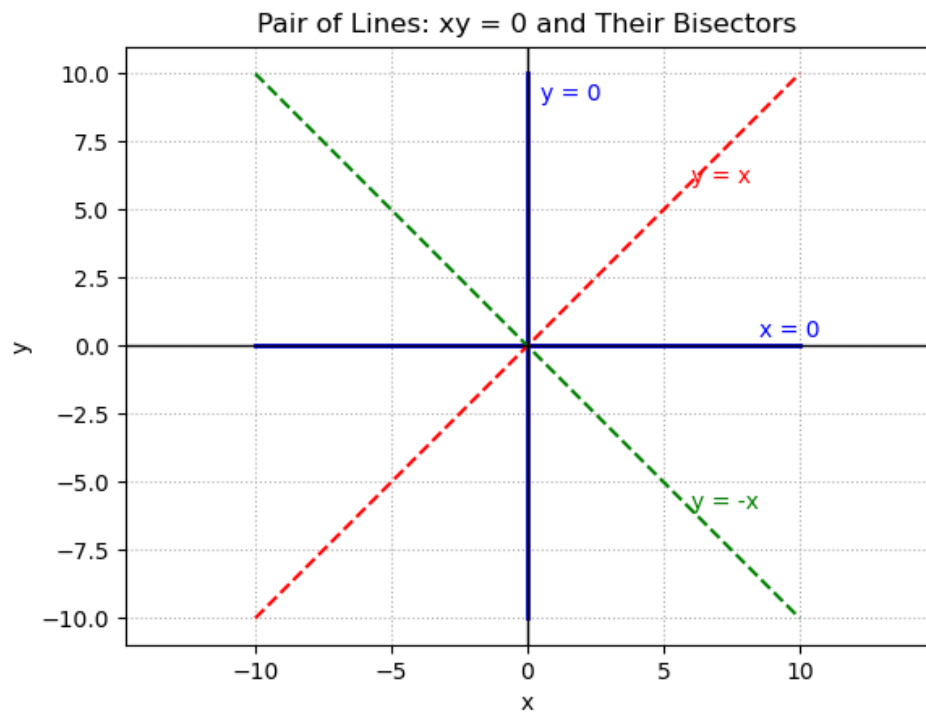


Figure 1: Lines